Generalization and metric divergence in overparameterized Conformer-CTC ASR

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Abstract

We study generalization in overparameterized Conformer-CTC ASR trained on LibriSpeech and find that CTC loss and word error rate respond differently to increasing model capacity. In a depth sweep over $L \in \{1, \ldots, 16\}$ Conformer blocks (14 M to 105 M parameters), val loss rises after models begin to interpolate while val WER keeps improving monotonically, indicating that the two metrics operate in different regimes past the interpolation threshold. No double-descent peak appears in WER under clean labels within a 100 k step budget. A subset ablation across 5k–460k utterances shows that larger models generalize better than smaller ones in every data regime, but the fixed step budget confounds direct data-scale comparison (small subsets complete up to 320 effective epochs vs. fewer than four for the full corpus).

1 Introduction

In overparameterized Conformer-CTC [1, 2] ASR, CTC loss and word error rate diverge past the interpolation threshold: val loss rises as models begin to interpolate, while val WER continues to improve. This metric split is the headline finding of this work, and it suggests that the standard cross-entropy generalization gap is not a reliable proxy for downstream recognition quality in CTC-based systems.

Classical statistical learning theory predicts generalization should degrade beyond the interpolation threshold, but modern overparameterized models frequently improve further, a phenomenon described by the double-descent risk curve [3, 4]. That phenomenon has been documented mainly in image classification; its behavior in sequence-to-sequence tasks such as ASR is less studied. Here we investigate it in a Conformer-CTC system trained on LibriSpeech with a depth sweep over $L \in \{1, \ldots, 16\}$ blocks and a subset ablation across data scales.

The interpolation threshold is defined as $\mathcal{L}_{\text{train}} < 0.01$ throughout, following Belkin et al. [3].

2 Literature review

The double-descent risk curve was formalized by Belkin et al. [3], who argued that the classical bias-variance picture must be extended to account for overparameterized models that interpolate their training data. Nakkiran et al. [4] documented the phenomenon at scale, identifying model-wise, sample-wise, and epoch-wise variants across a range of modern architectures and training conditions. A closely related result due to Zhang et al. [5] established that deep networks can memorize randomly labeled data while still generalizing on natural data, undermining capacity-counting arguments and motivating the need for sharper complexity measures. Together, these results suggest that near-interpolation behavior is organized by effective model complexity and training regime rather than by raw parameter count alone.

Formal generalization bounds for neural networks have been studied through Rademacher complexity and related tools. The foundational framework due to Bartlett and Mendelson [6] develops symmetrization and contraction results that connect function-class complexity to uniform convergence bounds for bounded losses. Subsequent work by Golowich et al. [7] found that norm-based bounds can be made essentially size-independent under layerwise norm constraints, and Bartlett et al. [8] extended this line to spectral-norm complexity for deep networks. These results share a common message: parameter count alone is not the right complexity scale, and architecture-aware norms give tighter and more informative bounds. In the sequence-model setting, Lotfi et al. [9] obtained the first non-vacuous PAC-Bayes bounds for large language models through model compression and prediction smoothing, suggesting that similar ideas may transfer to structured sequence losses.

Connectionist Temporal Classification (CTC), introduced by Graves et al. [1], trains a sequence model by marginalizing the training objective over all monotone alignments between the acoustic input and the output transcript. This latent-alignment structure means that low training loss need not imply low transcript-level error, and introduces a gap between the optimized objective and the true transcript risk that is not present in standard classification. The Conformer architecture of Gulati et al. [2] augments transformer self-attention with a local depthwise convolution module in each block, combining global and local context modeling. Each Conformer block also contains a dense normalization stack of five LayerNorm operations and one BatchNorm; this structure partially absorbs weight perturbations before they reach the output, which has implications for sharpness-based complexity proxies.

3 Experimental setup

3.1 Model

All experiments use a Conformer-CTC architecture [2]. Each Conformer block contains a feed-forward module, multi-head self-attention, a convolution module with BatchNorm, and a second feed-forward module, connected with pre-norm residual connections and five layer-normalization operations per block. The CTC head is a single linear projection onto the vocabulary. The capacity dimension varied is the number of Conformer blocks $L \in \{1, 2, 3, 4, 6, 8, 12, 16\}$, with hidden dimension $d = 512$ and $h = 8$ attention heads fixed. Parameter counts range from 14 M at $L = 1$ to 105 M at $L = 16$.

3.2 Data

Models are trained on LibriSpeech `train-clean-360` and `train-clean-100` combined (≈460 h, 460 k utterances). The `dev-clean` split serves as the validation set and is evaluated at every checkpoint during training; it is not used for any hyperparameter selection. `test-clean` is held out entirely during training and used only for final evaluation of each completed run. Audio is represented as 80-dimensional log-mel spectrograms ($n_{\text{fft}} = 400$, hop=160, window=400 at 16 kHz). Targets are SentencePiece unigram tokens with a vocabulary of 1 000 pieces. Utterance-level CMVN is applied by default unless stated otherwise.

For the subset ablation study (Section 5), training is restricted to the first $N \in \{5\text{k}, 50\text{k}, 150\text{k}\}$ utterances drawn from the combined corpus, with all other settings identical to the full-data runs.

3.3 Training

All runs use the Adam optimiser with a warm-up cosine learning rate schedule ($\text{LR}_{\max} = 10^{-4}$, $\text{LR}_{\min} = 10^{-6}$, 4 k warm-up steps), weight decay 10^{-6}, batch size 16, and dropout disabled. Training runs for 100 000 optimiser steps. Models are evaluated every 5 000 steps on a fixed subset of up to 500 batches drawn from each split. WER is computed by greedy CTC decoding. A comparison of three CMVN modes (none, global, utterance-level) on a fixed $L = 8$ model found no discernible effect on training dynamics or final val WER (max inter-condition difference: 0.18 pp, within inter-seed variance; see Appendix A). All experiments use utterance-level CMVN.

3.4 Logged metrics

Sharpness proxy. A one-sided random-perturbation estimate of loss curvature:

$$\widehat{S} \;=\; \mathbb{E}_\delta[\max(0,\, \mathcal{L}(\theta+\delta) - \mathcal{L}(\theta))]\,, \qquad \delta = \frac{\varepsilon_{\text{rel}}\,\|\theta\|_2}{\|\xi\|_2}\,\xi, \quad \xi \sim \mathcal{N}(0, I), \tag{1}$$

where $\varepsilon_{\text{rel}} = 10^{-3}$. The expectation is approximated over $N_b = 100$ batches and $N_p = 3$ independent perturbations per batch. This metric is logged as an auxiliary indicator only. The one-sided form retains only perturbations that increase the loss, avoiding contamination from flat or downhill directions. The dense normalization stack (5×LayerNorm + BatchNorm per block) attenuates the effect of weight perturbations at the output, making the proxy unreliable as an absolute curvature estimate for this architecture; a more robust measure (filter-normalized or SAM-based) will be considered for Stage B.

Relative sharpness.

$$\widehat{S}_{\text{rel}} \;=\; \frac{\widehat{S}}{\mathcal{L}_{\text{test}} + \epsilon}, \qquad \epsilon = 10^{-12}. \tag{2}$$

Weight norm.

$$\|\theta\|_W = \sqrt{\sum_{p\in\theta} \|p\|_2^2}. \tag{3}$$

Generalization gap.

$$\Delta = \mathcal{L}_{\text{test}} - \mathcal{L}_{\text{train-eval}}. \tag{4}$$

$\mathcal{L}_{\text{train-eval}}$ is evaluated on a fixed random subset of the training set at each checkpoint, decoupled from the running training loss.

4 Results: capacity sweep

We study whether overparameterized ASR models exhibit a double-descent generalization curve, with test error rising near the interpolation threshold and then decreasing as capacity grows further. The capacity curve is characterized under clean training labels, varying depth L while holding $d = 512$ fixed. Two random seeds are used per configuration; all training settings follow Section 3.3.

At step 100 k, models with $L \geq 8$ satisfy $\mathcal{L}_{\text{train}} \leq 0.009$ and thus meet the interpolation criterion; smaller models retain non-trivial training loss.

4.1 Capacity vs. generalization

Test WER decreases monotonically with model size across all evaluated configurations (Table 1, Figures 1–2). Test loss follows the same overall trend but has a slight non-monotonicity at $L = 4$ (0.692 vs. 0.680 for $L = 3$) before resuming its decrease. No double-descent peak is visible in WER under clean labels within the 100 k step training budget. The generalization gap (Table 1) is largest for the non-interpolating models ($L \leq 2$) and shows a slight non-monotonicity at $L = 4$ (0.660 vs. 0.592 at $L = 3$) before decreasing for larger models.

Table 1: Results on the full training set (mean over 2 seeds). All loss and WER values are at step 100 k.

L	Params	Train WER	Test WER	$\mathcal{L}_{\text{train}}$	$\mathcal{L}_{\text{test}}$	Gen. Gap	$\|\theta\|_W$
1	14 M	19.1%	39.0%	0.423	1.258	0.836	137
2	20 M	7.6%	23.0%	0.173	0.798	0.626	171
3	26 M	3.7%	17.7%	0.088	0.680	0.592	198
4	32 M	1.1%	15.9%	0.032	0.692	0.660	222
6	44 M	0.5%	13.3%	0.018	0.626	0.608	260
8	57 M	0.2%	12.1%	0.009	0.607	0.597	293
12	81 M	0.1%	11.0%	0.004	0.580	0.576	346
16	105 M	0.0%	10.3%	0.002	0.571	0.568	390

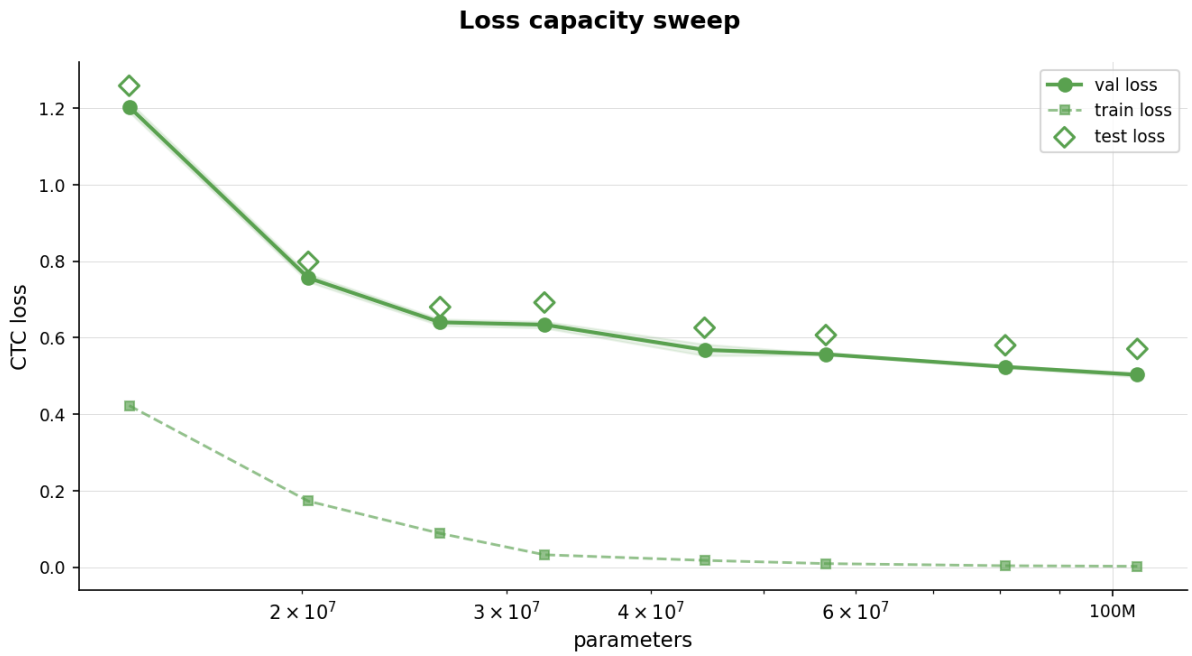

Figure 1: Train-eval loss and test loss at step 100 k as a function of model capacity (full training set). Train-eval loss decreases monotonically; test loss and val loss show a slight non-monotonicity at $L = 4$ before resuming their decrease. Shaded band: seed range.

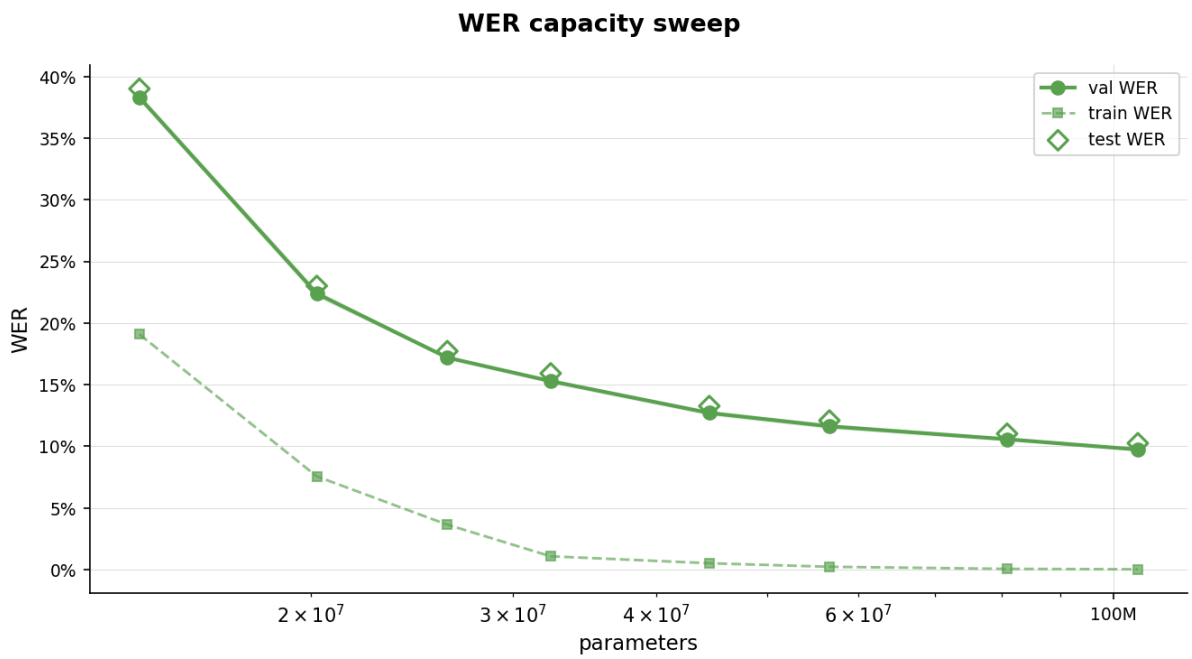

Figure 2: Train WER, val WER (dev-clean), and test WER (diamonds) vs. parameter count at step 100 k (full training set). Shaded band: seed range.

4.2 Epoch-wise trajectory

Val loss rises once models begin to interpolate (Figures 3–4), consistent with classical overfitting. Val WER diverges from this trend: larger models keep reducing WER long after training loss reaches near zero. Shallower models ($L = 4$, $L = 8$) reach a val WER minimum at around step 75 k and rise slightly; deeper models ($L \geq 12$) keep improving through step 100 k. Greedy CTC decoding is a threshold operation, so argmax improvements can coexist with rising cross-entropy.

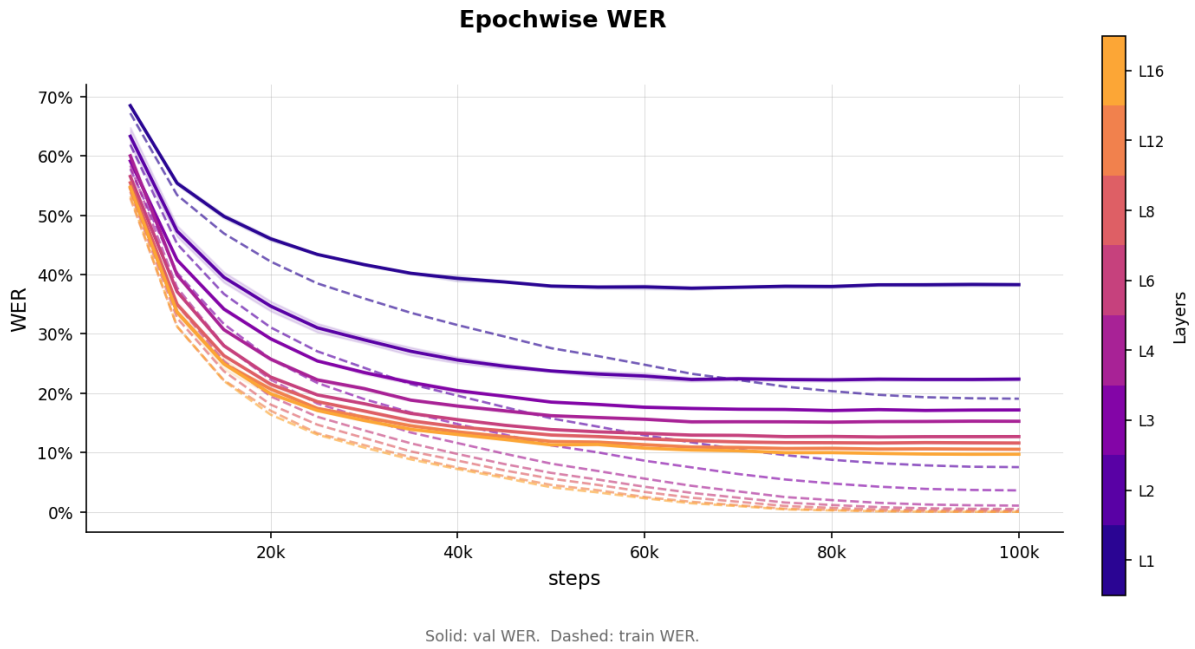

Figure 3: Val WER (solid) and train WER (dashed) over training steps for all capacity levels (full training set). Shaded band: seed range.

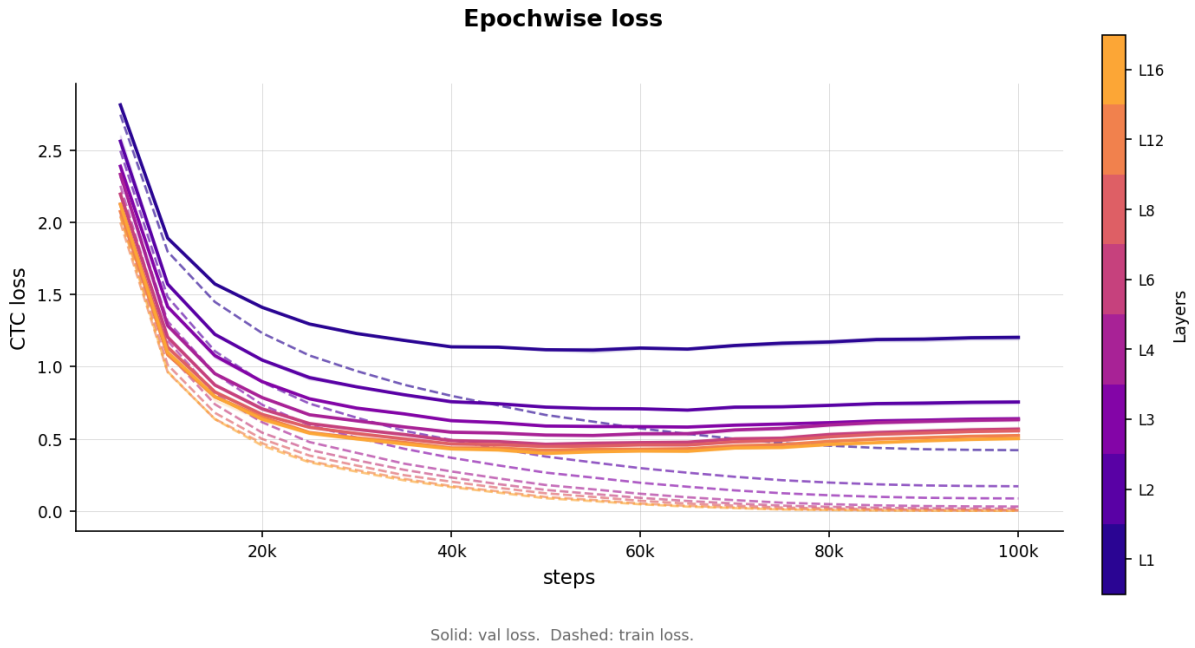

Figure 4: Val loss (solid) and train-eval loss (dashed) over training steps. Val loss rises after interpolation while train loss approaches zero. Shaded band: seed range.

Inter-seed variation in final test WER is below 0.45 percentage points for all configurations. Relative sharpness ($\widehat{S}_{\mathrm{rel}}$) values are uniformly small ($\approx 10^{-4}$) with no consistent ordering by depth; given the proxy's known limitations with dense normalization stacks (Section 3.4), these values are not interpreted further. Training curves for sharpness and weight norm are in Appendix A.

5 Results: subset ablation

The capacity sweep was repeated with training corpora restricted to $N \in \{5\text{k}, 50\text{k}, 150\text{k}\}$ utterances, probing two complementary regimes: one where every model overfits the training corpus, and one where smaller models underfit. Together these locate the experimental setup relative to the interpolation threshold across a range of data scales.

All subset runs use the same fixed budget of 100 000 optimiser steps, regardless of corpus size. Smaller subsets therefore complete many more effective passes over the training data: the 5k-utterance runs cycle through the training set roughly 320 times within that budget, whereas the full-dataset runs complete fewer than four epochs. Some of the generalization difference observed across corpus sizes may partly reflect deeper optimization rather than a purely data-scale effect. Disentangling the two factors requires running subset experiments to convergence or at matched epochs rather than matched steps; this convergence study is planned for Stage B.

5.1 Small subsets

With 5k and 50k training utterances, every model reaches near-zero training loss and training WER before the end of training (Table 2), including $L = 1$ (14 M parameters). All models are in the overparameterization regime. Yet val WER decreases monotonically from $L = 1$ to $L = 16$: larger models generalize better despite memorizing the same training set.

Table 2: Final step (100 k) metrics for the small subsets. All models reach train WER = 0% and $\mathcal{L}_{\text{train}} \approx 0$.

		5k utterances			50k utterances		
L	Params	Val WER	Test WER	Gen. Gap	Val WER	Test WER	Gen. Gap
1	14 M	77.2%	77.2%	13.03	54.1%	54.7%	4.14
2	20 M	71.4%	71.4%	10.77	33.6%	34.3%	2.44
3	26 M	67.7%	67.6%	9.91	25.6%	25.9%	1.85
4	32 M	66.7%	66.9%	9.34	21.7%	22.3%	1.61
6	44 M	63.1%	63.5%	8.75	18.4%	18.7%	1.37
8	57 M	61.6%	61.6%	8.30	16.7%	17.1%	1.25
12	81 M	59.3%	59.4%	7.69	15.0%	15.5%	1.13
16	105 M	56.4%	56.3%	7.29	14.4%	14.5%	1.03

No double-descent peak appears: $L = 1$ (14 M parameters) fully memorizes 5k or 50k utterances, so no model underfits these corpora. Only the descending branch is accessible; observing the peak would require either much smaller models or an even smaller training corpus. Training trajectory plots for the small subsets are in Appendix A.

5.2 Large subsets

With 150k or the full ∼460k utterances, small models ($L = 1$, $L = 2$) fail to fit the training data within 100 k steps (train WER 19%, 8%), while larger models progressively approach near-zero training loss. The 150k results are closely consistent with the full-dataset sweep; both are summarized in Table 3. The generalization gap (Table 3) follows a similar pattern to the full-data sweep: a slight increase at $L = 4$ for the 150k subset (0.634 vs. 0.597 at $L = 3$), followed by a decrease for larger models.

Table 3: Final step (100 k) metrics for the large subsets.

		150k utterances				Full (≈460k)			
L	Params	Train WER	Test WER	$\mathcal{L}_{\text{train}}$	Gen. Gap	Train WER	Test WER	$\mathcal{L}_{\text{train}}$	Gen. Gap
1	14 M	18.9%	39.1%	0.420	0.837	19.1%	39.0%	0.423	0.836
2	20 M	7.6%	23.0%	0.173	0.623	7.6%	23.0%	0.173	0.626
3	26 M	3.6%	17.7%	0.088	0.597	3.7%	17.7%	0.088	0.592
4	32 M	1.4%	15.9%	0.040	0.634	1.1%	15.9%	0.032	0.660
6	44 M	0.5%	13.3%	0.018	0.609	0.5%	13.3%	0.018	0.608
8	57 M	0.2%	12.2%	0.009	0.592	0.2%	12.1%	0.009	0.597
12	81 M	0.1%	10.9%	0.004	0.576	0.1%	11.0%	0.004	0.576
16	105 M	0.0%	10.1%	0.003	0.557	0.0%	10.3%	0.002	0.568

These runs sit on the underfitting side of the capacity curve. No double-descent peak appears here either: $L = 16$ reaches near-interpolation ($\mathcal{L}_{\text{train}} = 0.002$), but no generalization penalty is observed within 100 k steps. The model family spans the transition from underfitting to near-interpolation but does not clearly enter the overparameterization regime within the current training budget.

5.3 Cross-subset overview

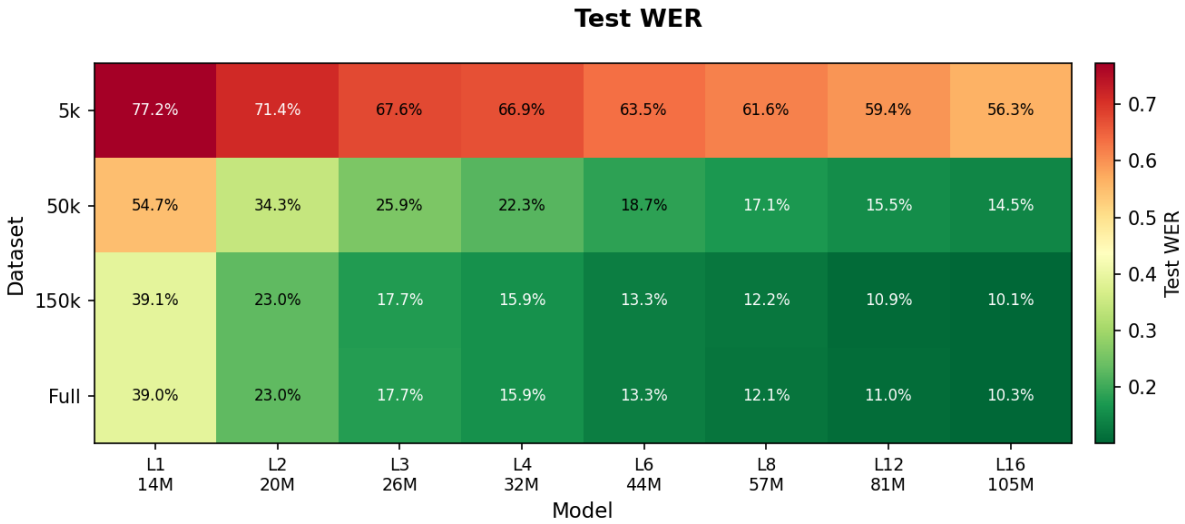

Figure 5: Test WER at step 100 k across all subset sizes (rows) and model capacities (columns). Larger models consistently achieve lower test WER within each row.

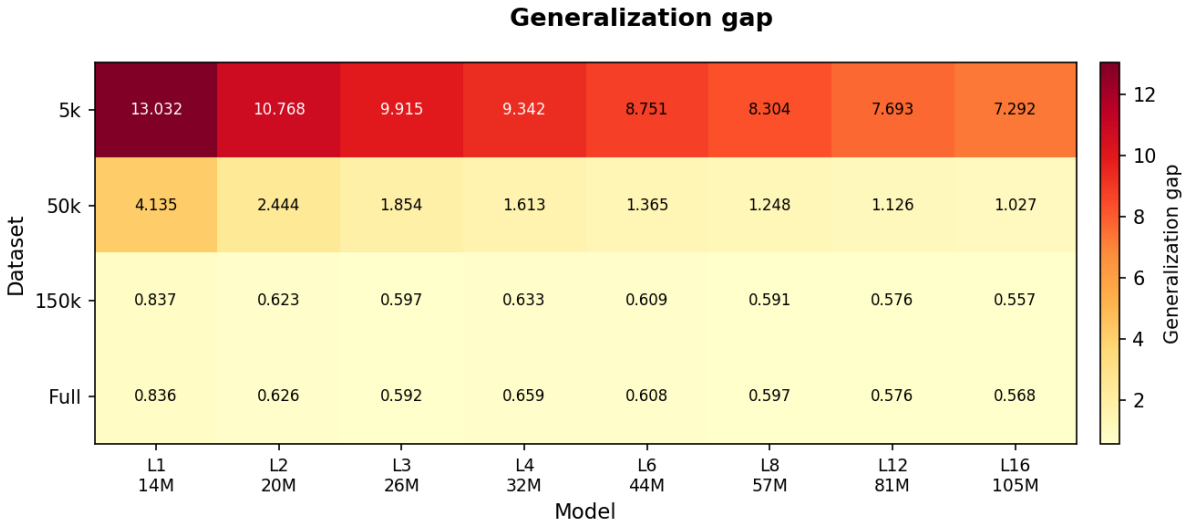

Figure 6: Generalization gap at step 100 k across all subset sizes and model capacities. For small subsets all models are in the overparameterization regime: the gap remains elevated but decreases monotonically with model size. For large subsets, small models have a large gap from underfitting; larger models show a decreasing gap as they approach interpolation.

Figures 5–6 summarize both regimes. Small subsets place all models on the overparameterization branch: every model overfits and generalization improves with capacity. Large subsets span the transition from underfitting to near-interpolation. Reproducing the double-descent peak requires a dataset that places some models in underfitting and larger ones in clear overfitting; the current capacity range does not achieve this within 100 k steps.

6 Limitations

The capacity sweep covers $L \in \{1, \ldots, 16\}$ Conformer blocks (up to 105 M parameters). Extending the sweep to larger models would require substantially more GPU memory and training time than available for Stage A; runs beyond $L = 16$ are planned once additional compute is secured. As a result, the right branch of the expected double-descent curve is not yet directly observable in the current data.

The subset ablation uses a fixed budget of 100 000 optimiser steps for all corpus sizes. At batch size 16, the 5k-utterance runs complete roughly 320 effective passes over the training data (100k steps $\times$ 16 / 5k utterances), while the full-corpus runs complete fewer than four. The smaller subsets therefore benefit from substantially more optimization time, which confounds the comparison with a pure data-scale effect. Comparing at convergence rather than at matched steps is needed to isolate the two factors; this is deferred to Stage B.

The Conformer architecture contains a dense normalization stack: five LayerNorm operations and one BatchNorm per block. These normalizations partially absorb weight perturbations before they reach the output, which makes the one-sided sharpness proxy unreliable as an absolute curvature estimate. The proxy is reported for completeness and as a Stage B baseline, but conclusions about loss-landscape geometry cannot be drawn from it for this architecture without a filter-normalized or SAM-based alternative.

7 Discussion

The full-data capacity sweep finds monotonically decreasing test WER with model depth and no double-descent peak within 100 k steps under clean labels. Double descent requires a training set near the interpolation threshold, or label noise [3]. On 460 h of clean speech, $L = 16$ reaches near-interpolation ($\mathcal{L}_{\text{train}} = 0.002$), but the dataset may be too large relative to the evaluated capacity range for a generalization peak to appear within this budget. Val WER keeps improving long after training loss approaches zero (Figures 3–4). For $L \geq 12$ the val WER curve has not turned by step 100 k, suggesting these models may still be improving within the current budget.

The subset ablation reveals two complementary regimes. On small corpora (5k and 50k), every model interpolates the training set but larger models generalize better, consistent with the beneficial overparameterization effect. On large corpora (150k and full), small models underfit and do not reach the interpolation threshold. No double-descent peak appears in either condition. The metric split between CTC loss and WER is a recurring pattern: val loss rises after interpolation while val WER keeps improving, reflecting the gap between cross-entropy and argmax decoding. Both quantities are reported throughout; their separate behavior is itself informative about the structure of the overparameterized regime in CTC ASR.

8 Future directions

Three experimental directions remain for completing the double-descent characterization. A training set of approximately 75 k–100 k utterances could place the interpolation threshold between $L = 4$ and $L = 8$, making both branches of the capacity curve accessible in a single sweep. Stage B label noise at 5%, 10%, and 20% corruption may raise the effective interpolation threshold for a given model size, potentially revealing the double-descent peak within the current model family on the full training set. Models beyond $L = 16$, or runs beyond 100 k steps, would push the largest models into clear overfitting and make the overparameterization regime unambiguous.

On the theoretical side, a formal generalization account for CTC-based ASR is being developed. The planned framework decomposes the population risk into three terms: the empirical training loss, a generalization gap between training and expected loss, and an annotation-bias term that captures the mismatch between the CTC objective, which marginalizes over all valid alignments, and the true transcript-level risk. Two proof routes are under investigation: a PAC-Bayes route that uses utterance-level compression and an unbounded-loss transformation following the approach of Lotfi et al. [9], and a Rademacher complexity route built on a logit-Lipschitz lemma for CTC and a norm-based complexity decomposition of the Conformer encoder following the framework of Bartlett and Mendelson [6] and Bartlett et al. [8]. These are planned contributions for a later stage of the project.

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A Additional figures

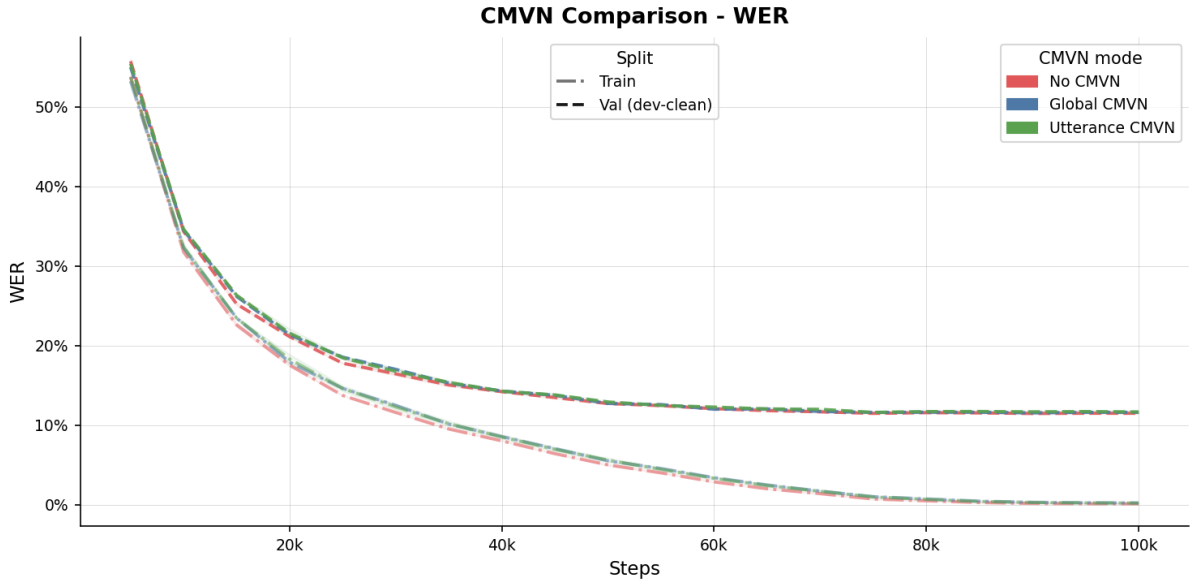

Figure 7: WER over training steps by CMVN mode.

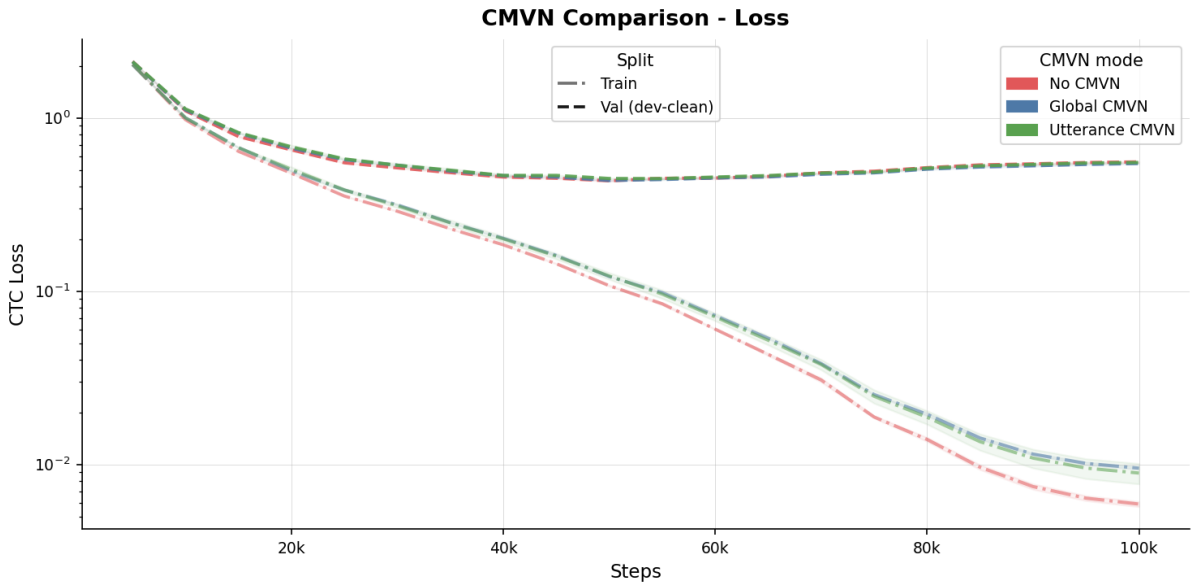

Figure 8: CTC loss over training steps by CMVN mode (log scale).

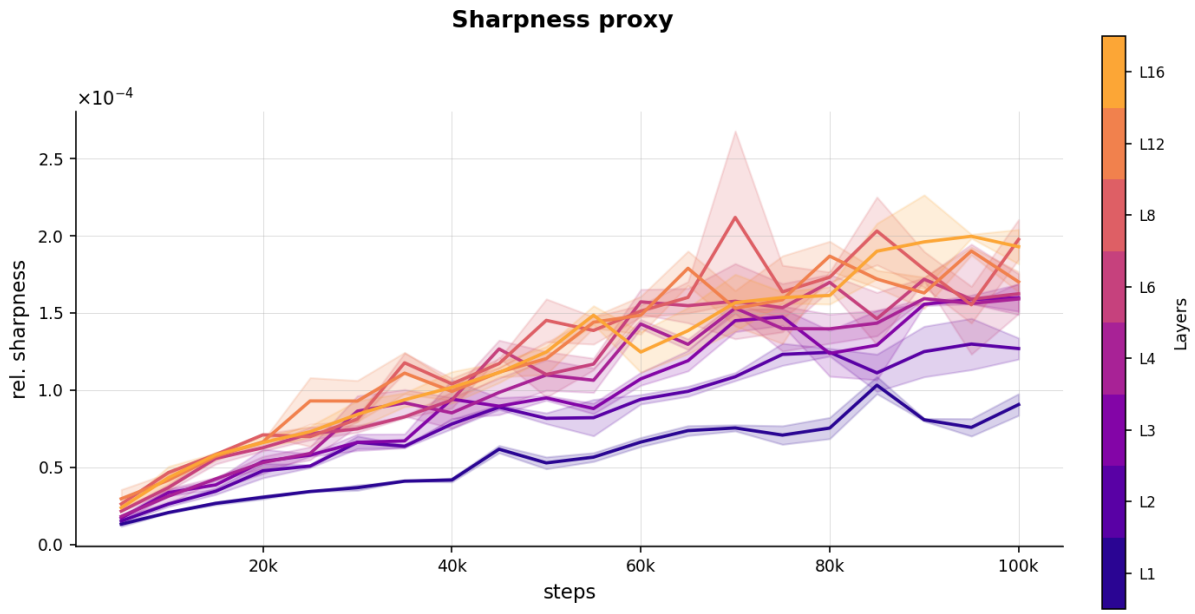

Figure 9: Relative sharpness proxy ($\widehat{S}_{\text{rel}}$, Eq. 2) over training steps (full training set). Shaded band: seed range.

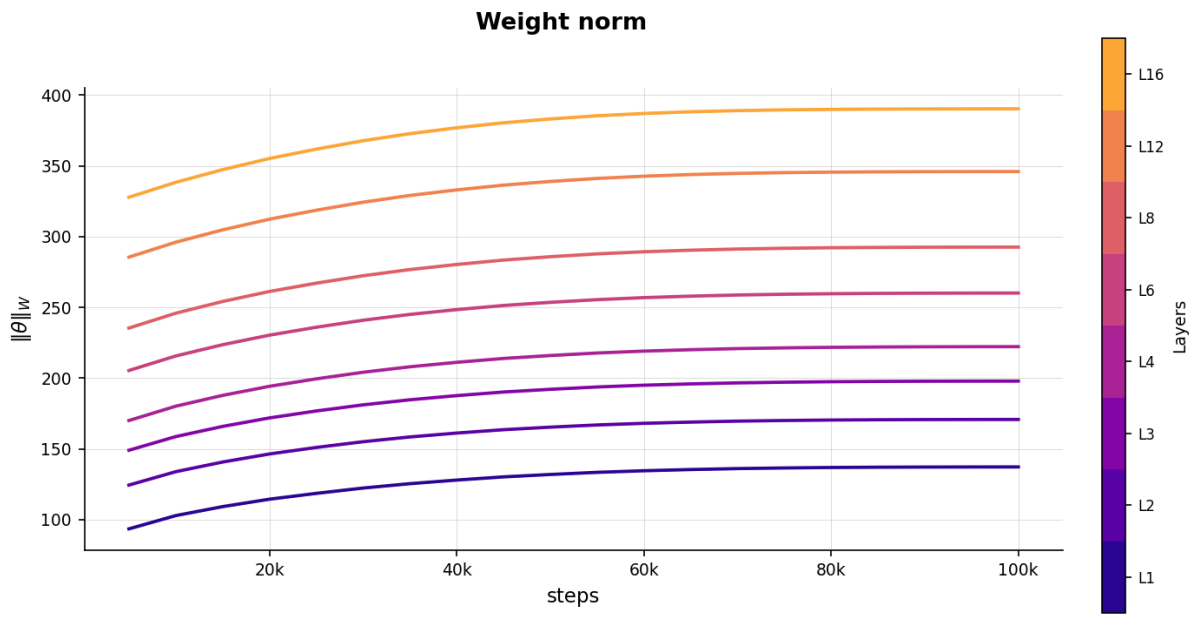

Figure 10: Weight norm $\|\theta\|_W$ over training steps (full training set). Shaded band: seed range.

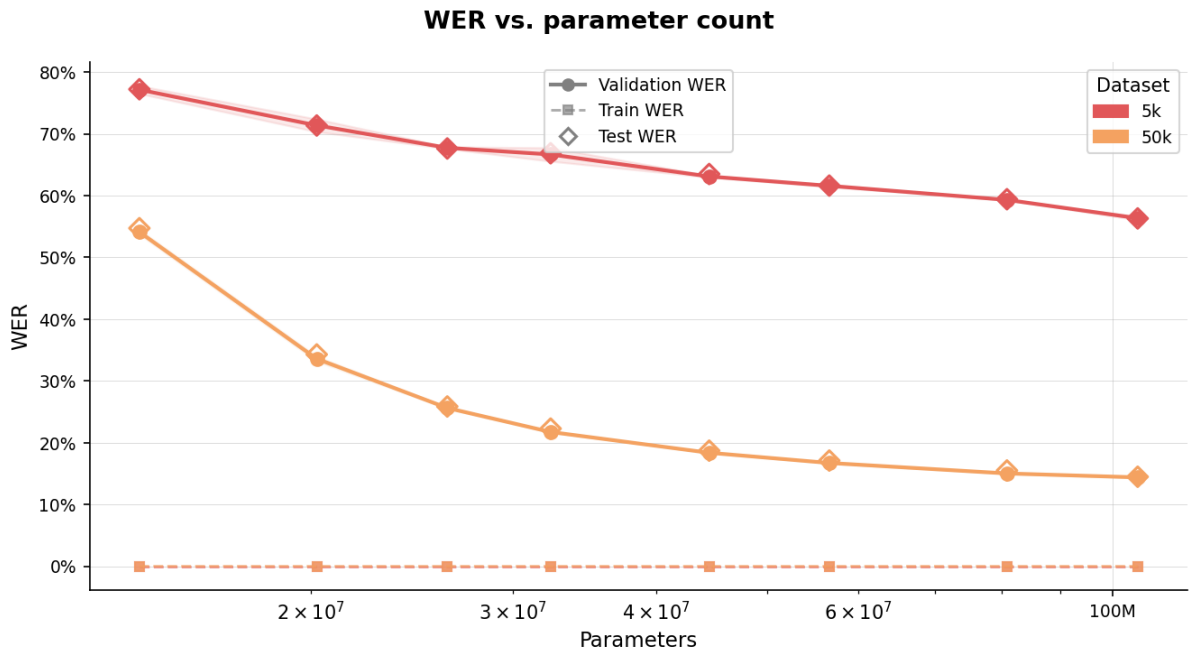

Figure 11: WER vs. parameter count at step 100 k for the small subsets (5k and 50k utterances). All models reach train WER = 0%; larger models nonetheless achieve lower val and test WER.

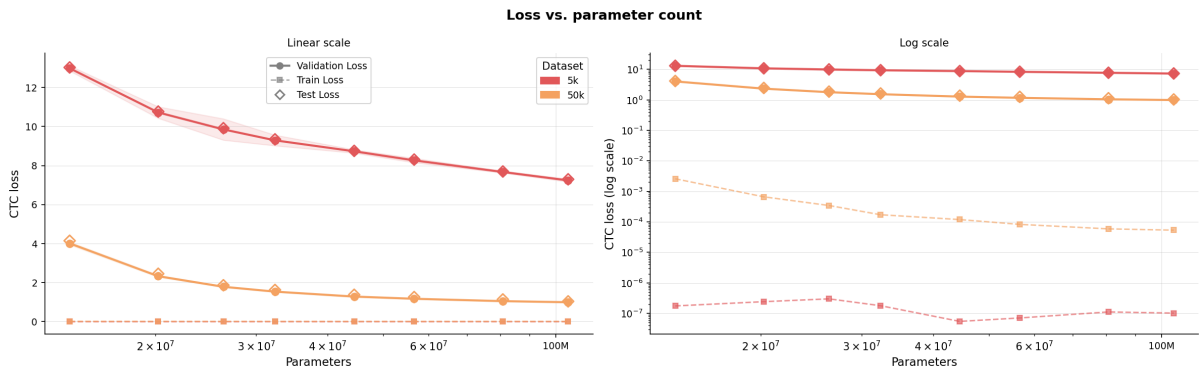

Figure 12: Loss vs. parameter count at step 100 k for the small subsets. Train-eval loss is near zero for all models. Shaded band: seed range.

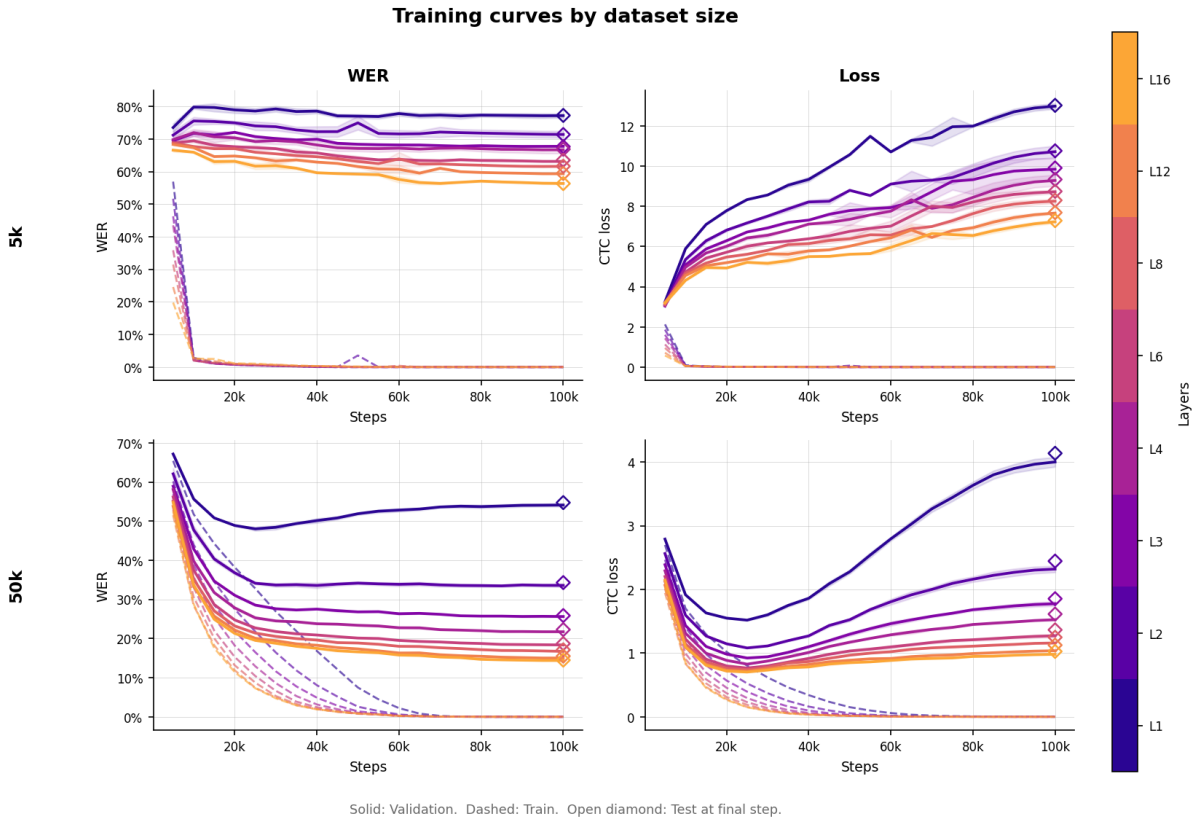

Figure 13: Training curves (WER, left; loss, right) for the small subsets. Each row is one subset size; color indicates model depth. Solid: validation. Dashed: train. Open diamond: test WER at final step.

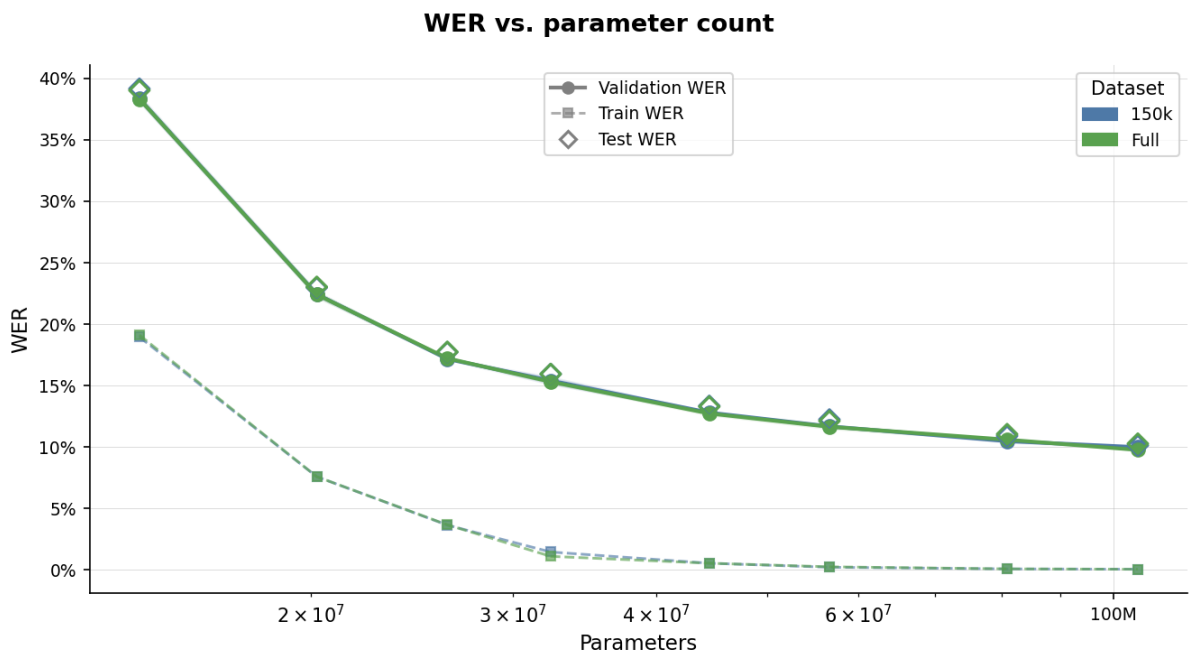

Figure 14: WER vs. parameter count at step 100 k for the large subsets (150k utterances and full dataset). Small models ($L \leq 2$) remain in the underfitting regime; larger models approach near-zero training loss. Shaded band: seed range.

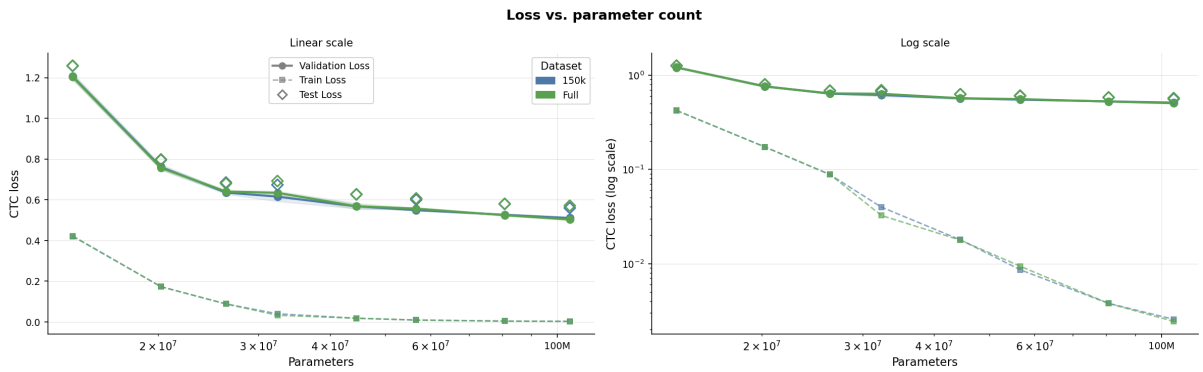

Figure 15: Loss vs. parameter count at step 100 k for the large subsets. Train-eval loss decreases with depth but remains non-trivial for small models. Shaded band: seed range.

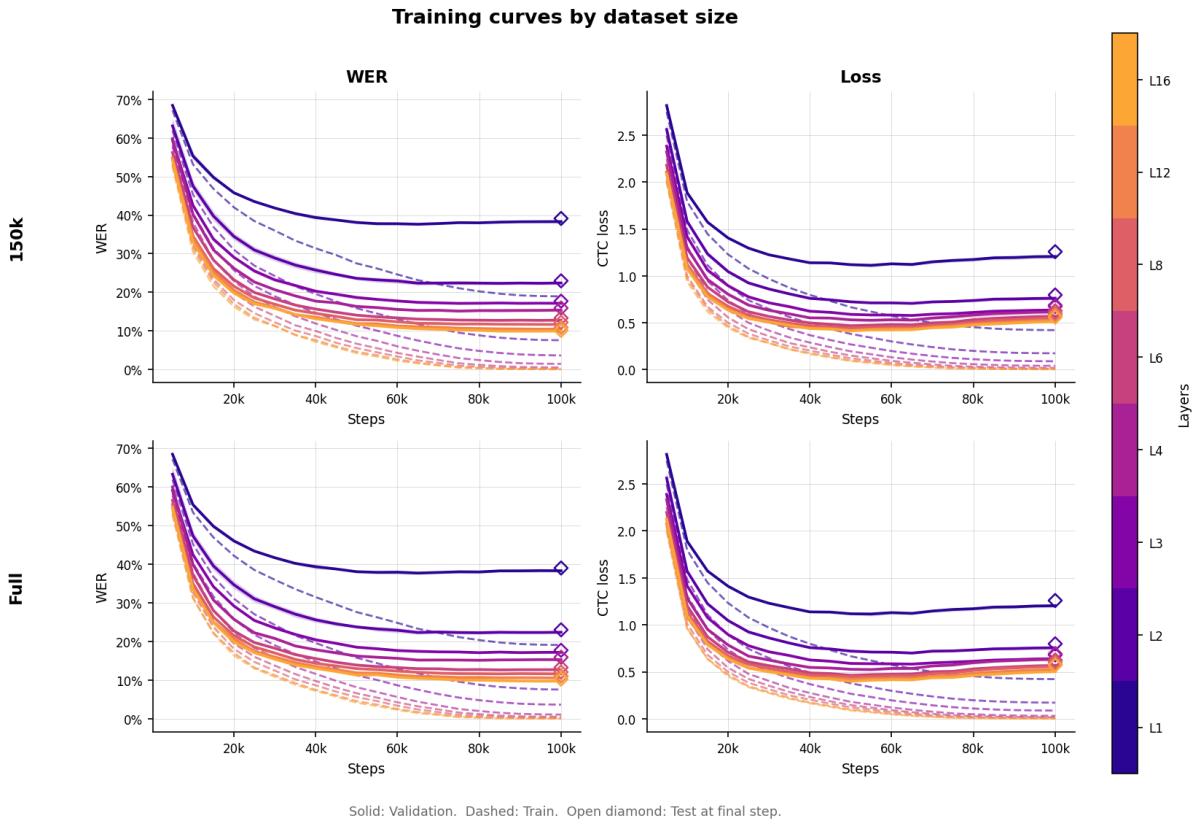

Figure 16: Training curves (WER, left; loss, right) for the large subsets. Each row is one subset size; color indicates model depth. Solid: validation. Dashed: train. Open diamond: test WER at final step.